SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE:ITA0604 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:100**

Annexure A

Analytical Problem Solving

Code of Conduct:

I D.RITHVIK {192312119}________ (Name / Reg No) certify that this submission in my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ***Total Marks Awarded***

Criteria	*Conceptual Understanding*	*Accuracy of Answers*	*Application of Knowledge*	*Completeness of Response*	*Clarity & Presentation*	*Total*
Weightage	*25%*	*25%*	*20%*	*15%*	*15%*	
Q1						
Q2						
Q3						

Annexure A

Short Answer Test

11. A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are 0.5 and 0.3, and the bias is –0.4. For a customer with advertisement exposure = 2 and income level = 1, compute the net input to the perceptron and apply the step activation function. Determine

whether the customer makes a purchase or not.
. (5 Marks)

12. In a neural network, a neuron is trained using the delta learning rule. The initial weight is 0.6, the learning rate is 0.15, and the input is 2.5. The actual output is 0.7, while the target output is 1. Calculate the error, determine the weight update, and compute the new updated weight after one iteration. (5 Marks)

13. A classification model produces the following confusion matrix values: True Positives = 140, True Negatives = 180, False Positives = 25, and False Negatives = 35. Compute accuracy, precision, recall (sensitivity), specificity, and F1-score. Based on the results, analyze whether the model is suitable for applications such as fraud detection where false positives are costly.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

11. Perceptron Model

Given:

- Weight $w_1 = 0.5$(Advertisement Exposure)
- Weight $w_2 = 0.3$(Income Level)
- Bias $b = -0.4$
- Input:
  - $x_1 = 2$

- $x_2 = 1$

Step 1: Calculate Net Input

$Net\ Input\ = w_1x_1 + w_2x_2 + b\ = (0.5\times2) + (0.3\times1) + (-\ 0.4) = 1.0\ +\ 0.3\ -\ 0.4\ =\ 0.9$

Step 2: Apply Step Activation Function

Step Activation Function:

$$f(x) = \{1,\ x\geq0\ 0,\ x\ <\ 0$$

Since

$$0.9\ >\ 0$$

Output = **1**

Decision

Output = **1**, therefore **the customer will purchase the product.**

12. Delta Learning Rule

Given:

- Initial Weight $w\ =\ 0.6$
- Learning Rate $\eta\ =\ 0.15$
- Input $x\ =\ 2.5$
- Actual Output $y\ =\ 0.7$
- Target Output $t\ =\ 1$

Step 1: Calculate Error

$$Error\ =\ t\ -\ y\ =\ 1\ -\ 0.7\ =\ 0.3$$

Step 2: Calculate Weight Update

Delta Rule:

$$\Delta w\ =\ \eta\times(t\ -\ y)\times x\ =\ 0.15\times0.3\times2.5\ =\ 0.1125$$

Step 3: Calculate New Weight

$$w_{new}\ = w\ +\ \Delta w\ =\ 0.6\ +\ 0.1125\ =\ 0.7125$$

Final Answer

- **Error = 0.3**
- **Weight Update = 0.1125**
- **New Weight = 0.7125**

13. Confusion Matrix Performance Metrics

Given:

- True Positives (TP) = 140
- True Negatives (TN) = 180
- False Positives (FP) = 25
- False Negatives (FN) = 35

Step 1: Accuracy

$$Accuracy\ = \frac{TP+TN}{TP+TN+FP+FN} = \frac{140+180}{140+180+25+35} = \frac{320}{380} = 0.8421\ =\ 84.21\%$$

Step 2: Precision

$$Precision\ = \frac{TP}{TP+FP} = \frac{140}{140+25} = \frac{140}{165} = 0.8485\ =\ 84.85\%$$

Step 3: Recall (Sensitivity)

$$Recall = \frac{TP}{TP+FN} = \frac{140}{140+35} = \frac{140}{175} = 0.80 = 80\%$$

Step 4: Specificity

$$Specificity = \frac{TN}{TN+FP} = \frac{180}{180+25} = \frac{180}{205} = 0.8780 = 87.80\%$$

Step 5: F1-Score

$$F1 = \frac{2\times Precision\times Recall}{Precision+Recall} = \frac{2\times0.8485\times0.80}{0.8485+0.80} = \frac{1.3576}{1.6485} = 0.8237 = 82.37\%$$